Sampling variability

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

List the ID samples. Compute the statistic. State the sampling model.

Explain exact finite-population variability and distinguish it from unrelated observed classifier-rate changes.

Keep one habit: Sampling contract → all-six-pair ledger/native chart → classifier picture → model/replay probes → exact enumeration → independent new-value population.

MAIN LESSON · SECTION 1

See how different eligible samples change a statistic

> Hold population and procedure fixed while tracing every possible sample.

Learning goals

I can explain the **mechanism** and terminology of **sampling variability**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve units, repeated observations and controlled comparisons

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Confounders and controls. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **sampling variability** using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **sampling variability**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Define the population, sample rule and statistic before counting outcomes

> The probability belongs to this explicit sampling model.

Begin with four numbered cards

Imagine four numbered cards with values one, one, three and three. Choose two different card IDs without replacement. The mean depends on which cards were included even though the population and calculation never changed. Here every unordered ID pair is equally likely, so we can list all possibilities and compute exact probabilities. The equal values still belong to different cards.

These six possibilities are not six newly observed experiments.

Define sampling variability

Sampling variability means different eligible samples can give different estimates even under the same population. Separate this from procedure changes.

Complete known-population probability contract

Sampling-model item	Complete fictional finite-population rule
Known population	U1=1, U2=1, U3=3, U4=3; four distinct IDs

Sampling-model item	Complete fictional finite-population rule
Sample	Two different IDs, without replacement within a sample
Probability model	All six unordered ID pairs equally likely
Statistic	Arithmetic mean of the two selected values
All possible sample means	1, 2, 2, 2, 2, 3 in the declared pair order
Probability of mean 2	Four of six eligible pairs = 2/3
Average of all possible means	2, equal to this known population mean
Dependence and counting	Do not merge IDs with equal values; without-replacement draws are not independent
Separate picture	Different four-case classifier samples score 3/4 and 2/4; no sampling probability model supplied
Scope	Exact enumeration under the stated toy model, not a random experiment, confidence interval or real model evaluation

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace every ID pair and the resulting sampling distribution

> A single sample mean can differ from the known population mean.

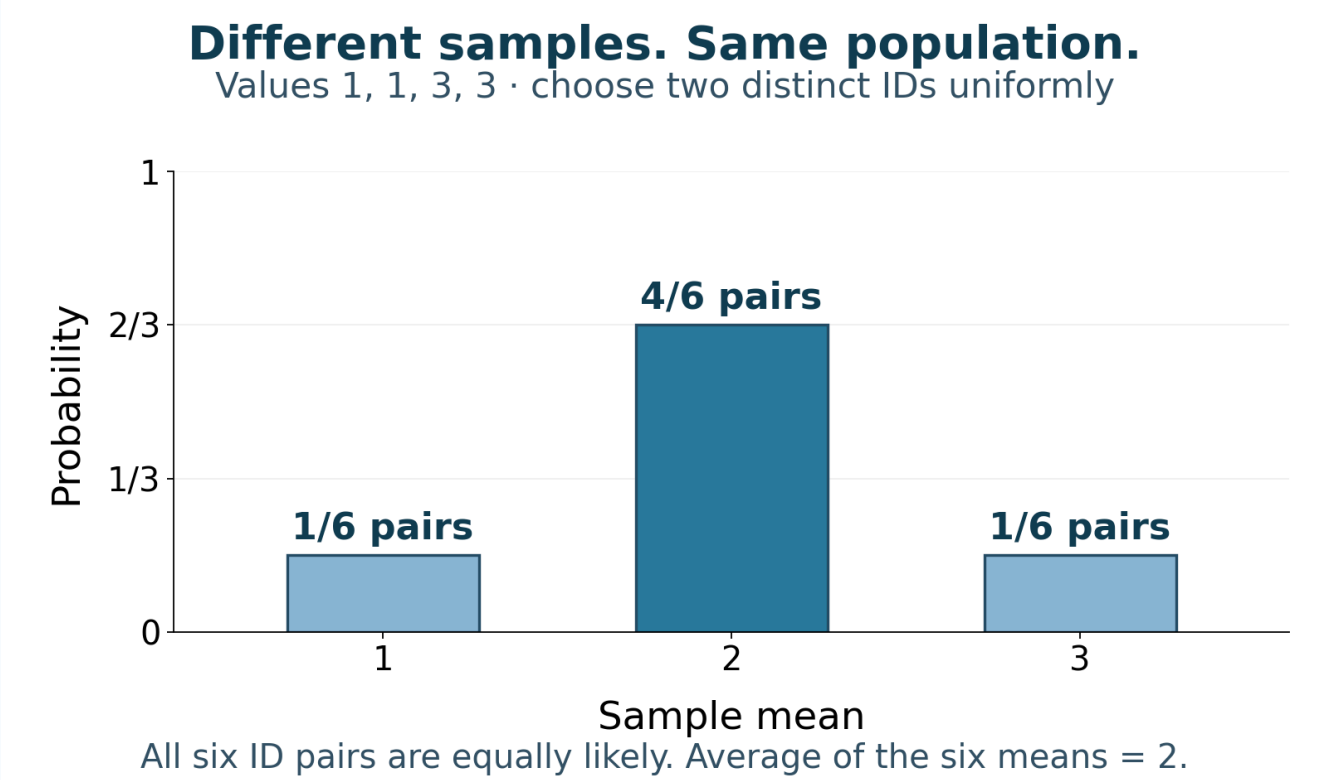
Original finite-population enumeration rule

Consider finite population values 1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size 2 **without replacement**. Their means are 1, 2, 2, 2, 2, 3.

Complete six-pair value/mean/probability ledger

Unordered ID pair	Selected values	Sample mean	Model probability
U1, U2	1, 1	1	1/6
U1, U3	1, 3	2	1/6
U1, U4	1, 3	2	1/6
U2, U3	1, 3	2	1/6
U2, U4	1, 3	2	1/6
U3, U4	3, 3	3	1/6
Summary	Six equally likely ID pairs	Average = (1+2+2+2+2+3)/6 = 2	P(mean 2) = 4/6 = 2/3

Read the exact distribution in the native chart



State the model before generalizing

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Sampling variability — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2.	Consider finite population values 1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size 2 without replacement.	Mean = (1+2+2+2+2+3)/6=2; mean 2 occurs 4/6=2/3. The population mean is 2, while individual samples differ.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict rerunning identical deterministic records

> Changed included cases and changed scoring are different explanations.

Current task and response

- Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the same fixed classifier cases scored twice

- Change: Evaluate the exact same fixed four cases twice with deterministic **prediction**s.

▼ Check after predicting

Prediction: The counts should agree unless some condition, data or scoring changed.

Reason: The sampling explanation no longer applies when the records are identical.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Read fixed-model sample differences without inventing a probability model

> Keep the exact card enumeration separate from the classifier rates.

Main complete-population problem and checked result

Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2.

Mean = (1+2+2+2+2+3)/6=2; mean 2 occurs 4/6=2/3. The population mean is 2, while individual samples differ.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate four-case sample inputs and scope

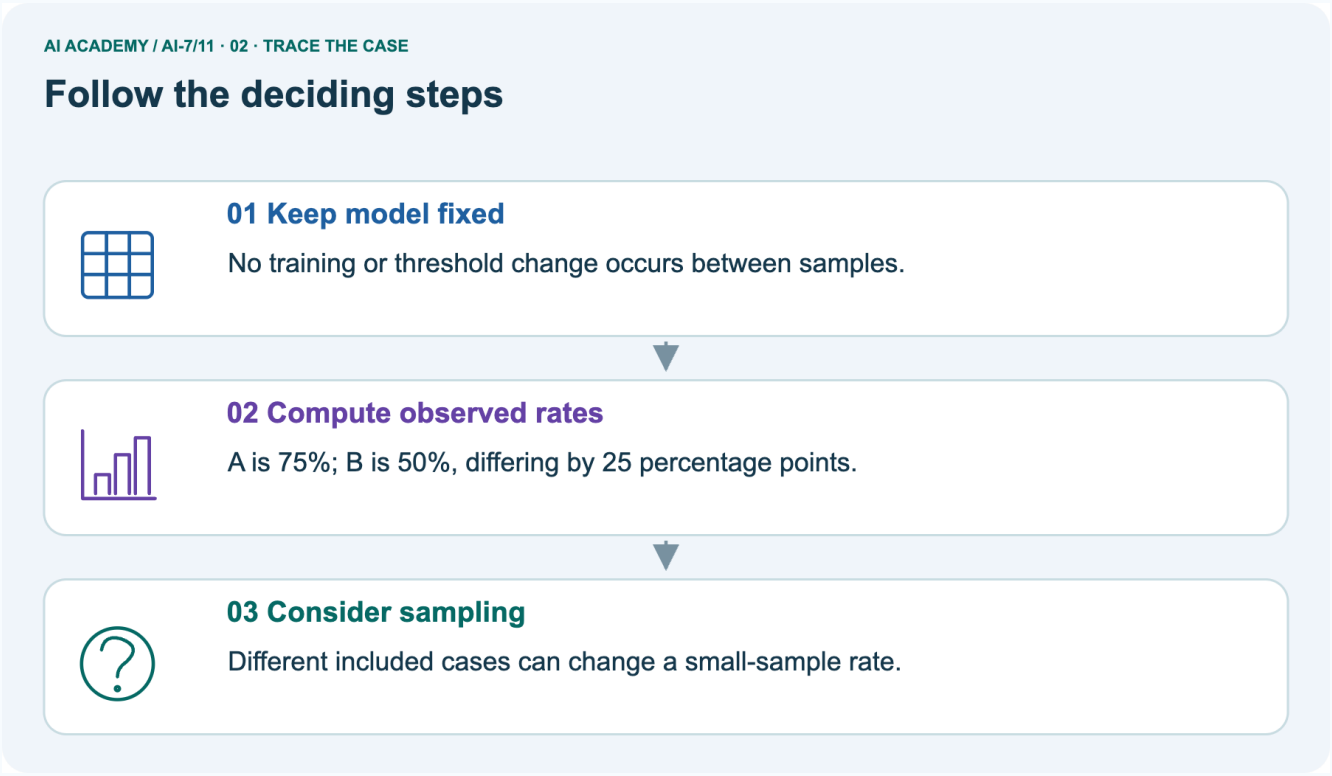
Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

Sample A has 3 correct of 4; sample B has 2 correct of 4. The model and rubric are unchanged.

The picture is a separate fixed-classifier example, not another draw from the four-card population. Sample A has three correct of four, or 75%; B has two of four, or 50%, a difference of 25 percentage points. Different included cases can change the observed rate without changing the classifier or rubric.

But no sampling design or probability model is supplied for these two samples, so do not borrow the main example’s 2/3 probability or invent a confidence interval. Repeating the same cases with deterministic fixed predictions and unchanged scoring should give the same count; investigate an unexplained mismatch rather than calling it new-sample variability.

Trace unchanged classifier, rates and possible sampling variation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep model fixed	No training or threshold change occurs between samples.
2. Compute observed rates	A is 75%; B is 50%, differing by 25 percentage points.
3. Consider sampling	Different included cases can change a small-sample rate.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Sampling variability means a statistic can vary because different units were observed. Do not automatically attribute every rate change to a model improvement or deterioration.

Compare what each example actually supplies

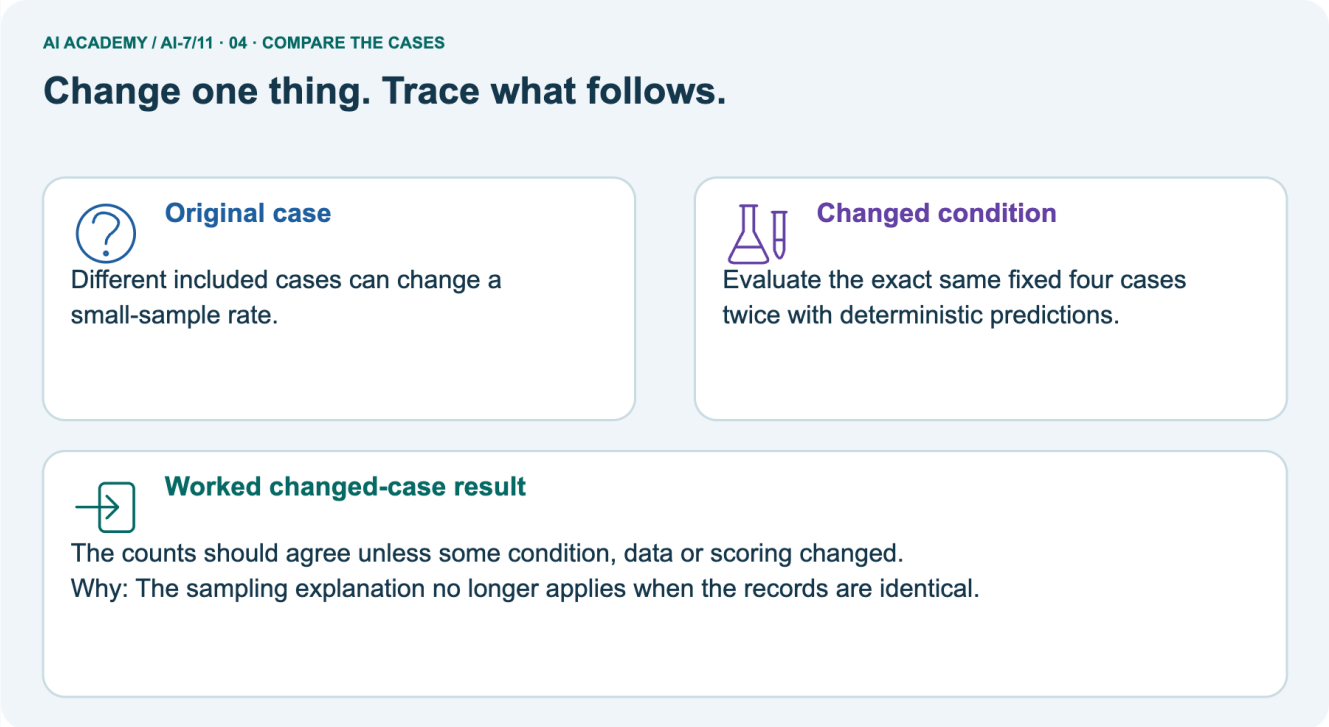
Evidence distinction	Main finite model	Separate classifier picture
Population/sample information	All four IDs/values and uniform pair selection known	Only two four-case counts supplied
Observed/possible quantities	Six possible means enumerated exactly	A 3/4 = 75%; B 2/4 = 50%
Computed probability	P(mean 2)=2/3 under stated model	No probability model supplied
What changes	Which two IDs enter a sample	Which four cases enter the scored sample

Evidence distinction	Main finite model	Separate classifier picture
Identical deterministic replay	Same chosen records give same mean	Same cases/predictions/scoring give same rate
Not established	Broader real-population inference	Cause, population precision or confidence interval

Predict deterministic replay on identical cases

- Change one thing: Evaluate the exact same fixed four cases twice with deterministic **prediction**s. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain missing probability model and repeated-record limits

- These two samples do not supply a probability model or confidence interval; none is invented.

- Read the labelled picture: Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

▼ Earlier wording, alternatives and picture transcripts

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
from itertools import combinations

def sample_means(population, size):
    if type(size) is not int or not 1 <= size <= len(population):
        raise ValueError('sample size')
    return [sum((population[i] for i in indices)) / size for indices in
combinations(range(len(population)), size)]

print(sample_means([1, 1, 3, 3], 2))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

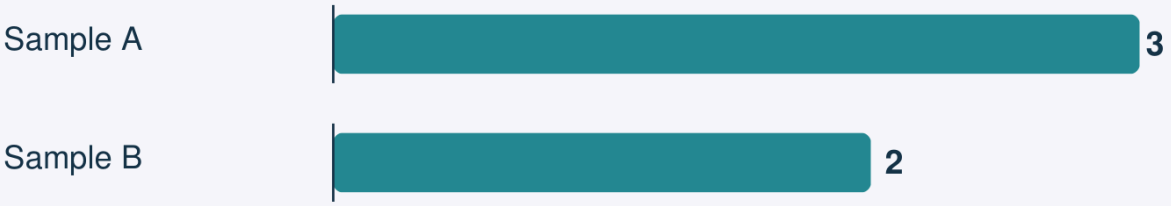
A pictorial worked case: Sampling variability

The facts we will use

AI ACADEMY / AI-7/11

Two distinct samples, fixed model

correct cases out of 4 in each sample



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/11

Sampling variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep model fixed

No training or threshold change occurs between samples.



2. Compute observed rates

A is 75%; B is 50%, differing by 25 percentage points.



3. Consider sampling

Different included cases can change a small-sample rate.

CHANGE ONE THING

Evaluate the exact same fixed four cases twice with deterministic predictions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit unit identity, selection probabilities and unexplained changes

> A sampling explanation requires stating what was sampled and how.

Core sampling qualification and failure analysis

This finite example is not a general normality guarantee or a model of a biased convenience sample.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete exact-model and replay probes

Sampling-variability probe	Expected interpretation
Equal values with distinct IDs	Remain separate eligible units
Choose two uniformly without replacement	Six equally likely unordered pairs for four IDs
Six sample possibilities listed	Not six independently observed experiments
A sample mean is 1 or 3	Compatible with fixed population mean 2 under this model
Four pairs have mean 2	Exact model probability 4/6, not a universal sampling rule
Selection rule changes	Recompute probabilities; do not reuse uniform-pair weights blindly
New classifier sample differs	May reflect changed included cases; not automatic model drift
Same deterministic cases score differently	Inspect data, code, settings and scoring changes

Sampling-variability probe	Expected interpretation
Only two observed rates supplied	No confidence interval or probability of the difference established

Guided repair and explanation

- A report omits this qualification: This finite example is not a general normality guarantee or a model of a biased convenience sample.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise listing samples by ID rather than unique value

> Use the declared sample rule to derive the denominator.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 trace

Use workbook W2 for the connected example "**Sampling variability**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for **Sampling variability**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

> **My explanation using a pictured step and the given facts**
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Reserve the selected zero/zero/four/four population

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run a complete finite-population enumeration with exact fractions

> Check the ID samples, statistic and probability weights together.

Current all-pairs sampling investigation

Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2. Show the intermediate steps and give the conclusion with its scope.

Runnable exact enumeration and scope checks

```
"""Exact finite-population enumeration under a declared uniform sampling model."""
from itertools import combinations
from fractions import Fraction
from collections import Counter

def enumerate_means(records,size):
 if type(records) is not list or not records:raise ValueError('nonempty finite population
required')
 if type(size) is not int or not 1<=size<=len(records):raise ValueError('valid exact
sample size required')
 ids=[]
 for record in records:
  if type(record) is not dict or set(record)!={'id','value'}:raise ValueError('exact
ID/value record required')
  if type(record['id']) is not str or not record['id'] or record['id'] in ids:raise
ValueError('distinct nonempty population IDs required')
  if type(record['value']) is not int:raise ValueError('exact integer values required for
this toy model')
  ids.append(record['id'])
 rows=[]
 for indices in combinations(range(len(records)),size):
  selected=[records[i] for i in indices]
  rows.append({'ids':tuple(r['id'] for r in selected),'values':tuple(r['value'] for r in
selected),
               'mean':Fraction(sum(r['value'] for r in selected),size)})
 return rows
population=[{'id':'U1','value':1},{'id':'U2','value':1},{'id':'U3','value':3},
{'id':'U4','value':3}]
```

```
rows=enumerate_means(population,2)
means=[r['mean'] for r in rows];frequencies=Counter(means)
# All six unordered ID pairs are equally likely under this explicitly declared model.
probabilities={value:Fraction(count,len(rows)) for value,count in
sorted(frequencies.items())}
average_of_means=sum(means)/len(means)
population_mean=Fraction(sum(r['value'] for r in population),len(population))
assert means==[1,2,2,2,2,3] and average_of_means==population_mean==2
assert probabilities==
{Fraction(1):Fraction(1,6),Fraction(2):Fraction(2,3),Fraction(3):Fraction(1,6)}
assert sum(probabilities.values())==1
assert len(enumerate_means(population,4))==1 and enumerate_means(population,4)[0]
['mean']==2
# Separate picture: observed counts only, no probability model inferred for these samples.
picture_a=Fraction(3,4);picture_b=Fraction(2,4)
assert 100*(picture_a-picture_b)==25
print('Six unordered ID samples:',[(r['ids'],r['values'],str(r['mean'])) for r in rows])
print('Uniform-model sample-mean probabilities:',{str(k):str(v) for k,v in
probabilities.items()})
print('Average sample mean:',average_of_means,'known finite-population
mean:',population_mean)
print('Separate picture rates:',float(100*picture_a),float(100*picture_b),'difference: 25
percentage points')
print('Equal values retain distinct IDs. Enumerated samples are possibilities, not six
independent observed experiments. Draws within a without-replacement sample are not
independent.')
print('No random draw, live classifier, population inference or confidence interval is
performed. The picture supplies no sampling probability model; it cannot borrow this
finite-model probability.')
```

Expected probabilities and no-random-study limits

1. Predict all six pair means and their frequencies before running or tracing. The exact-fraction code preserves distinct IDs with equal values and enumerates without replacement. It checks the average mean and known population mean both equal two, and the chance of mean two is four sixths. It also handles the full-population boundary as one possible sample, without changing the reserved independent values.

   Separate picture arithmetic gives 75% and 50%, not a derived sampling probability. No random draw, live classifier, six observed studies or confidence interval is produced. Keep actual local output distinct from a paper trace.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

```
from itertools import combinations

def sample_means(population, size):
    if type(size) is not int or not 1 <= size <= len(population):
        raise ValueError('sample size')
    return [sum((population[i] for i in indices)) / size for indices in
combinations(range(len(population)), size)]

print(sample_means([1, 1, 3, 3], 2))
```

Plan → try → check

Try alone: new values with the same distinct-ID sample rule

> Count all six pairs even when two population values are equal.

Complete selected independent task

Population values are 0, 0, 4, 4, with four distinct IDs. Choose two IDs uniformly without replacement. Enumerate all six unordered pairs, calculate their sample means and find their average.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

Population values are 0, 0, 4, 4, with four distinct IDs. Choose two IDs uniformly without replacement. Enumerate all six unordered pairs, calculate their sample means and find their average.

> **My independent procedure, result and limitation**
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save population IDs, selection rule and every sample statistic

> Keep exact model calculations separate from observed study evidence.

Project task and artifact

Add a labelled **sampling variability evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **sampling variability**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual arithmetic run or paper trace and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain why a statistic varies and which probability model supports the calculation

> Do not explain identical-record mismatches as new sampling.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **sampling variability**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Sampling variability	Changes in estimates across eligible samples
Sampling distribution	The distribution of an estimate under a sampling procedure
Without replacement	Selecting distinct units within a sample

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: This finite example is not a general normality guarantee or a model of a biased convenience sample.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Sampling variability — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2.	Consider finite population values 1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size 2 without replacement.	Mean = (1+2+2+2+2+3)/6=2; mean 2 occurs 4/6=2/3. The population mean is 2, while individual samples differ.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

By the end, I can	How I will show it
Explain the mechanism and terminology of sampling variability	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.
- Which labelled step matches this detail? Different included cases can change a small-sample rate.
- 1. Consider sampling
- 2. Keep model fixed
- 3. Compute observed rates
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Consider sampling Different included cases can change a small-sample rate. **Sampling variability** means a statistic can vary because different units were observed.

Do not automatically attribute every rate change to a model improvement or deterioration.

Changed-case question and optional hint

- Changed-case practice: Evaluate the exact same fixed four cases twice with deterministic **prediction**s. Which conclusion is supported by the supplied facts?
- 1. The counts should agree unless some condition, data or scoring changed.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The counts should agree unless some condition, data or scoring changed. The sampling explanation no longer applies when the records are identical.

Repair and retain the record

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the **mechanism** and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain.

Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Review Replication and reproduction (ai-7/5). Goal: Explain the **mechanism** and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

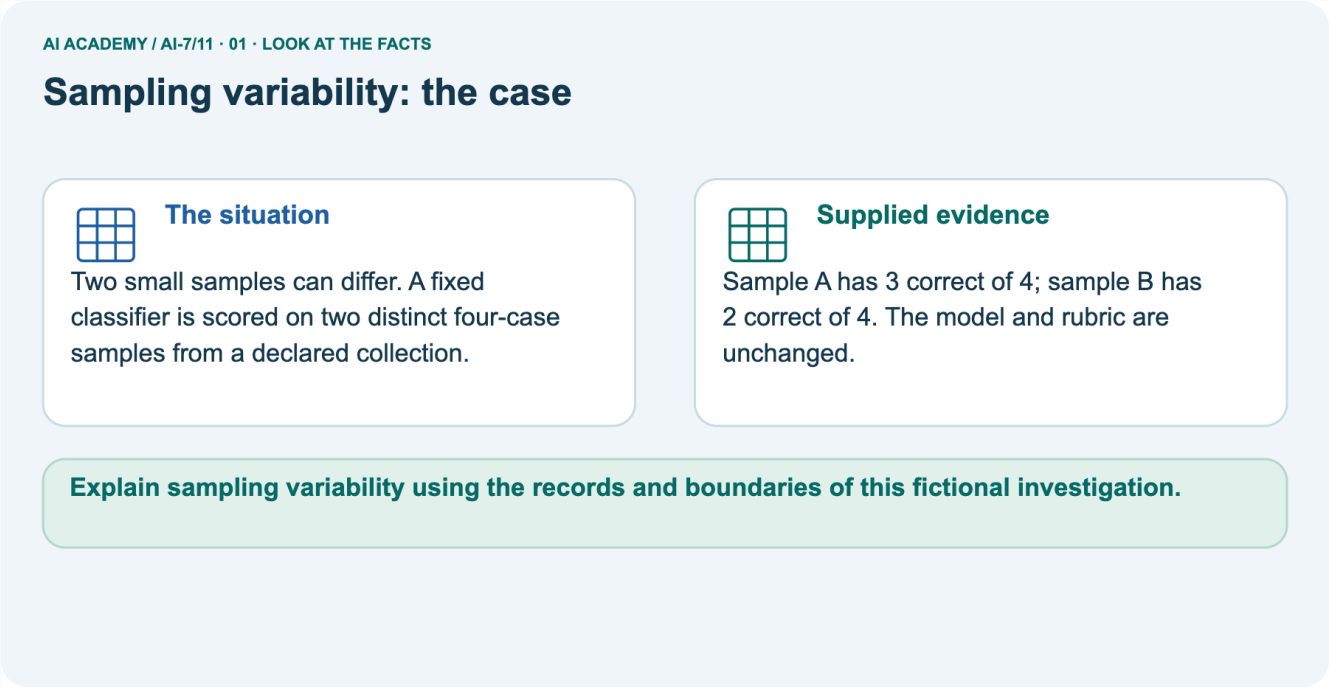
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Sampling variability: the case



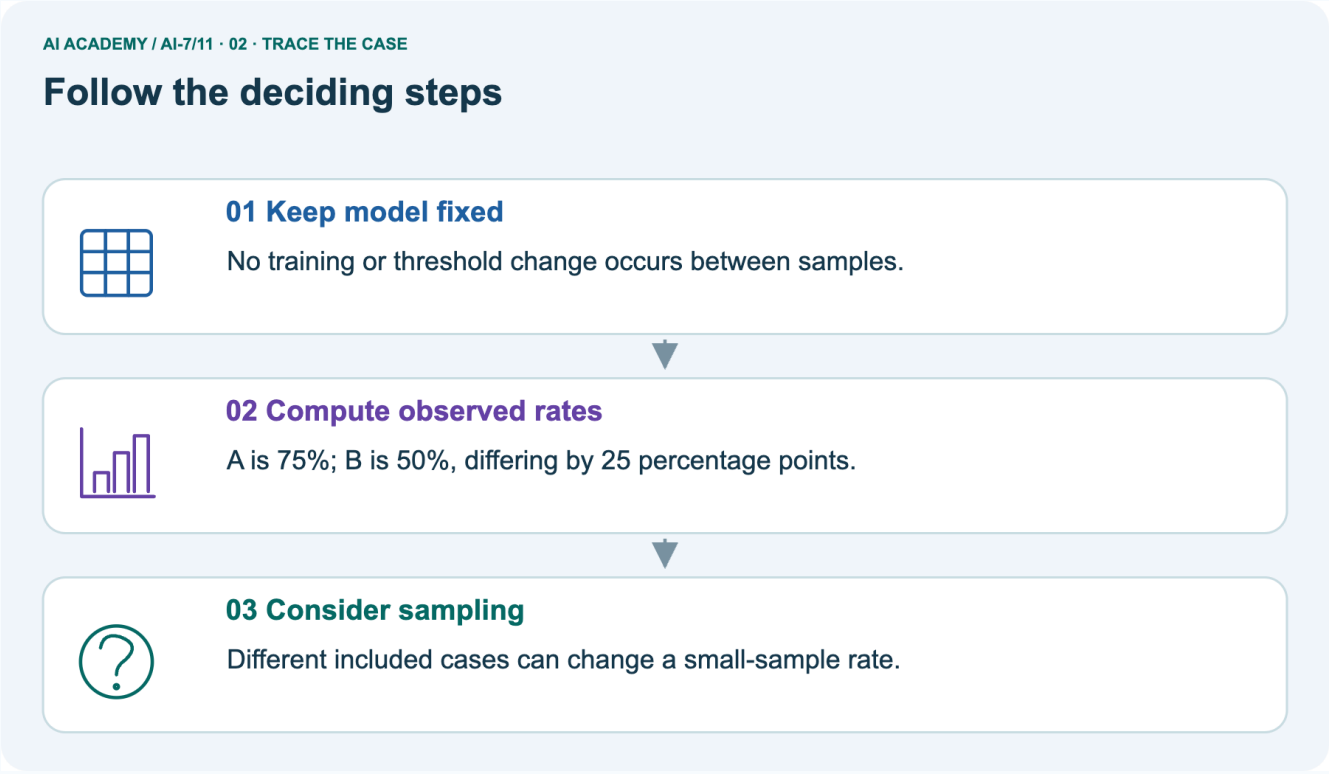
Follow the labelled picture with the matching explanation. Open the vector diagram

Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection. Supplied facts: Sample A has 3 correct of 4; sample B has 2 correct of 4.

The model and rubric are unchanged.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

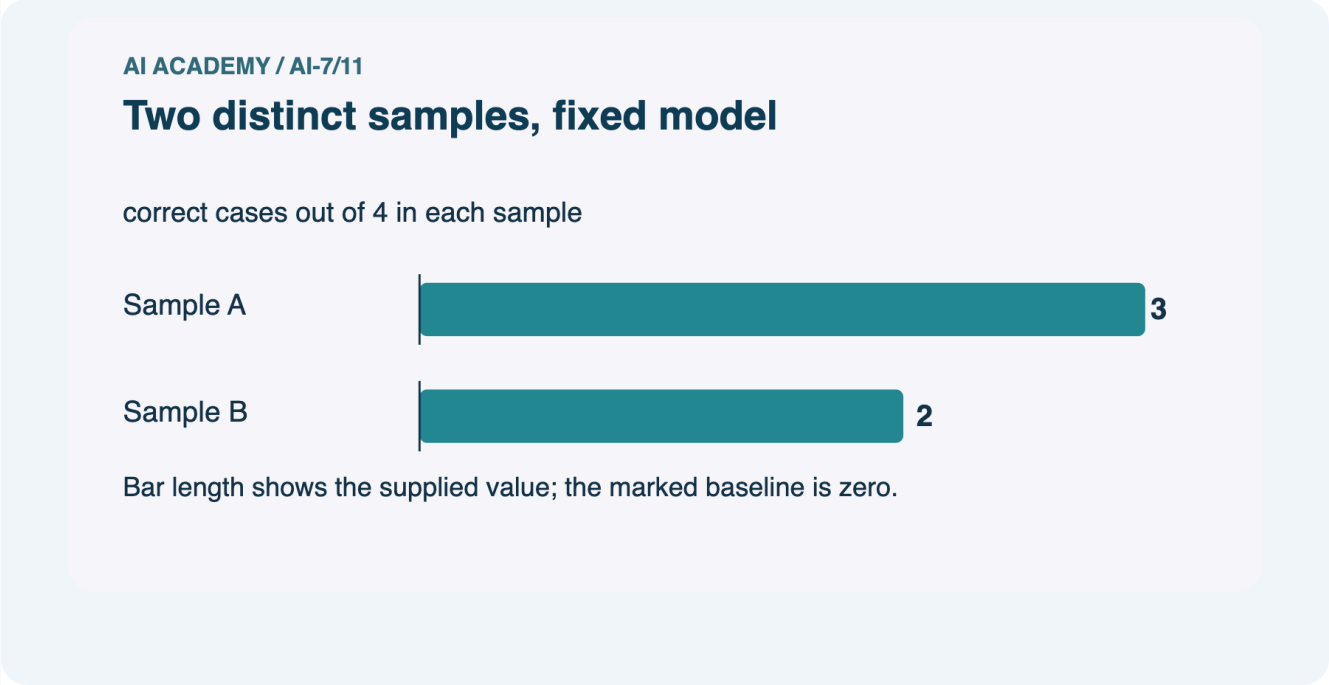


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep model fixed: No **training** or threshold change occurs between samples. → Compute observed rates: A is 75%; B is 50%, differing by 25 percentage points. → Consider sampling: Different included cases can change a small-sample rate.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two distinct samples, fixed model

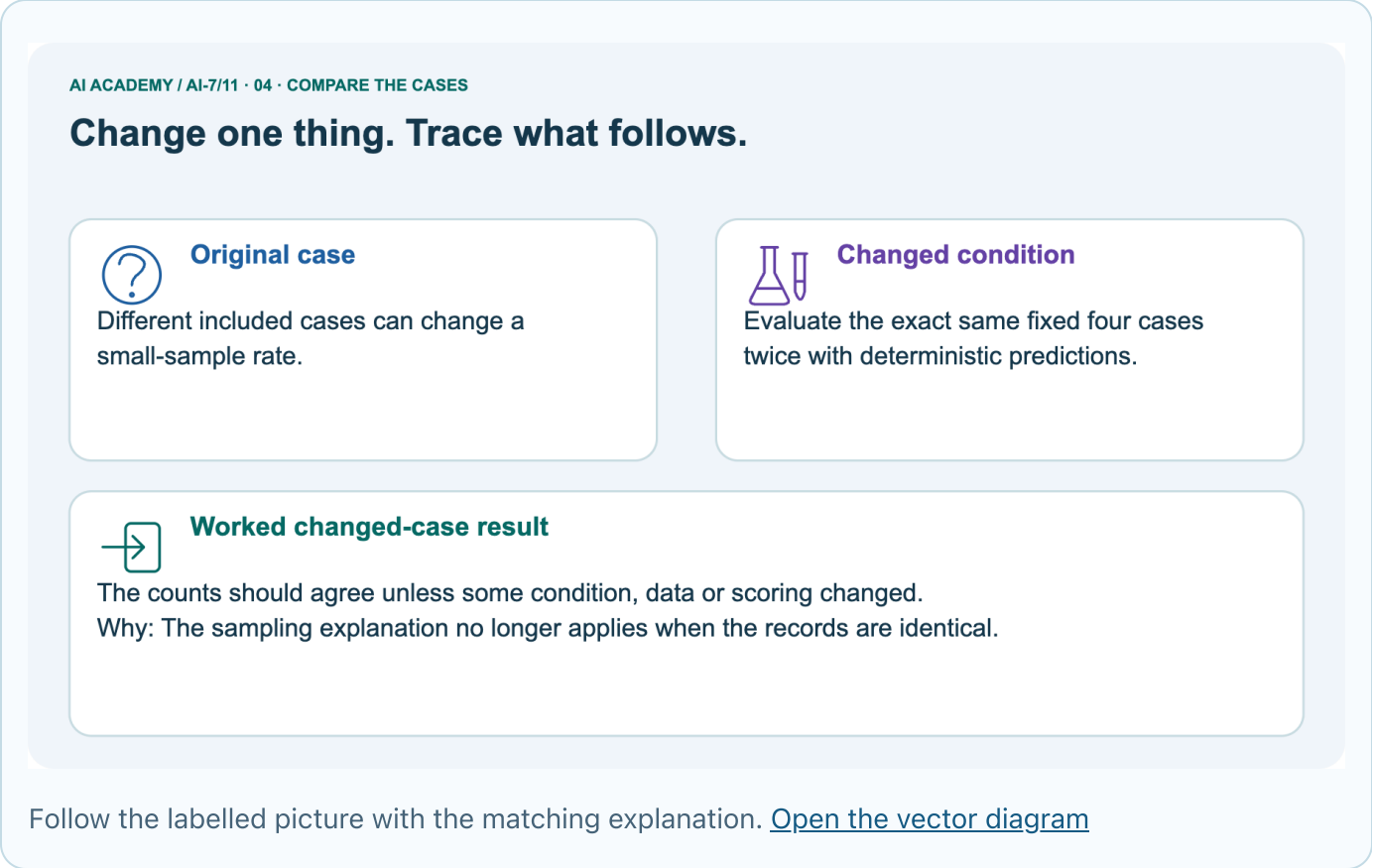


Follow the labelled picture with the matching explanation. Open the vector diagram

Two distinct samples, fixed model. Units: correct cases out of 4 in each sample. Sample A: 3. Sample B: 2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



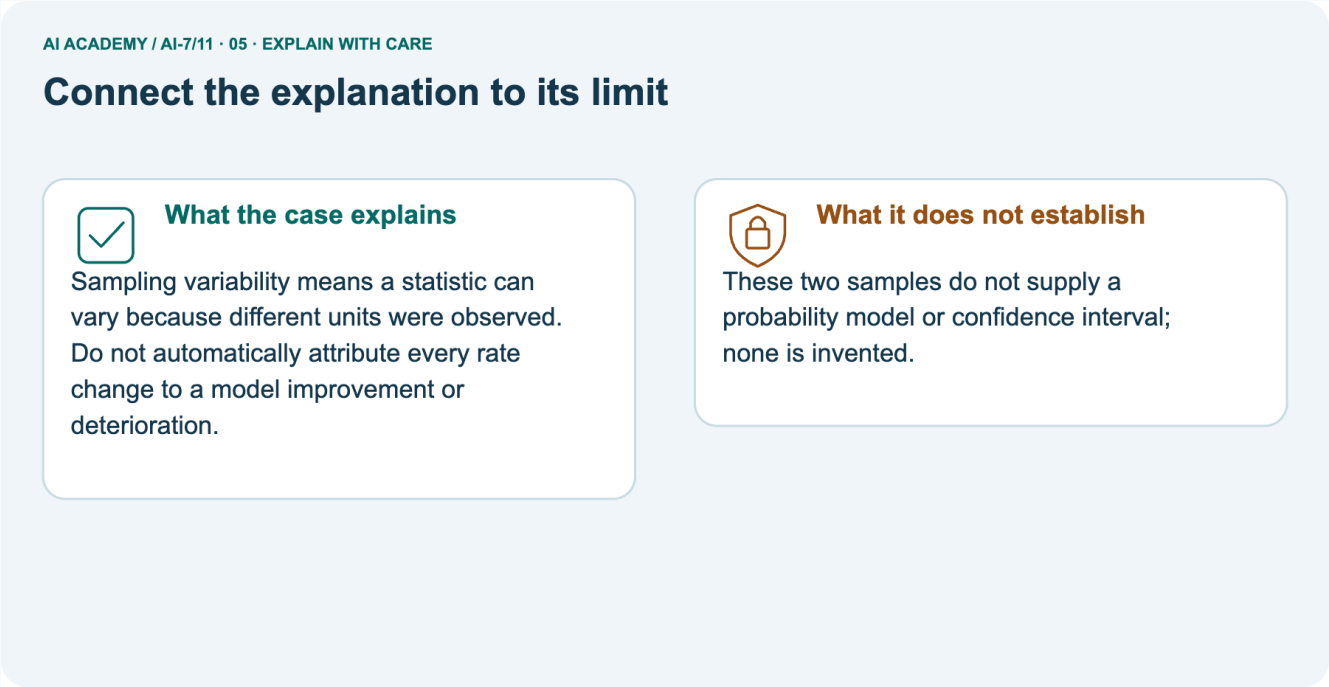
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Different included cases can change a small-sample rate. Changed condition: Evaluate the exact same fixed four cases twice with deterministic **prediction**s.

Expected result: The counts should agree unless some condition, data or scoring changed. Reason: The sampling explanation no longer applies when the records are identical.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



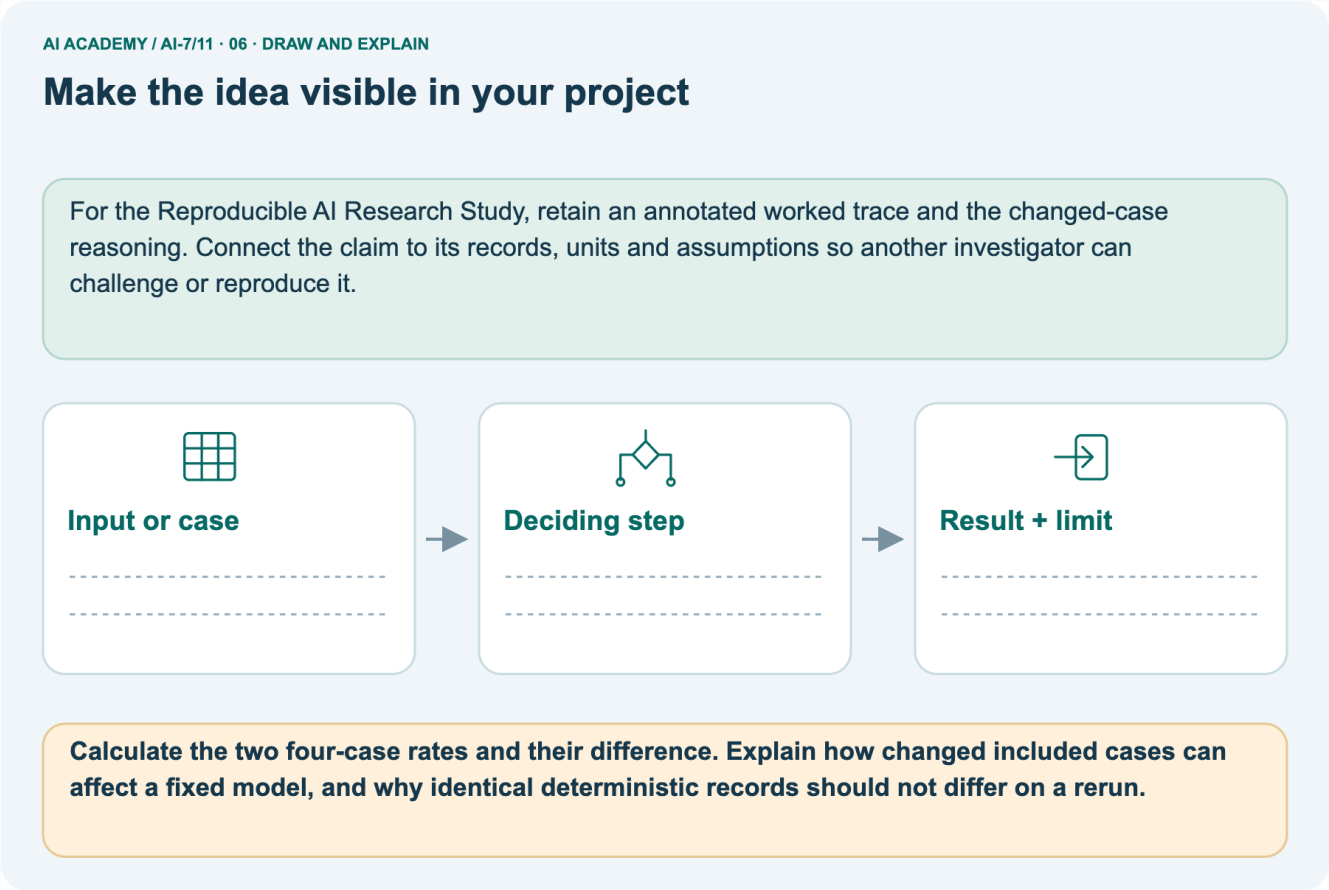
Follow the labelled picture with the matching explanation. Open the vector diagram

Sampling variability means a statistic can vary because different units were observed. Do not automatically attribute every rate change to a model improvement or deterioration.

Limit: These two samples do not supply a probability model or confidence interval; none is invented.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Sampling variability: the case

02. Follow the deciding steps

03. Two distinct samples, fixed model

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.